

NIH Biographical Sketch Common Form

Name: Shechter, David

Persistent Identifier (PID) of the Senior/Key Person: <https://orcid.org/0000-0001-9388-6004>

Position Title: Professor of Biochemistry

Organization and Location: Albert Einstein College of Medicine, Bronx, New York, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
Columbia University College of Physicians and Surgeons, New York, New York, United States	PhD	08/1997	05/2004	Cellular, Molecular, and Biophysical Studies
Cornell University, Ithaca, NY, United States	B.S.	08/1991	05/1995	Biological Sciences

Appointments and Positions

2021 - present	Professor of Biochemistry, Albert Einstein College of Medicine, Bronx, New York, United States
2015 - 2021	Associate Professor, Albert Einstein College of Medicine, Biochemistry, Bronx, New York, United States
2009 - 2015	Assistant Professor, Albert Einstein College of Medicine, Biochemistry, Bronx, New York, United States
2004 - 2009	Postdoctoral Fellow, Rockefeller University, Allis Lab, New York, New York, United States

Products

Products Closely Related to the Proposed Project

- Schurer A, Ilyas H, Maron M, Hegde S, Leyden M, Roy I, Hyka R, Dada L, Shabanowitz J, Hunt D, Angeles E, Morell V, Lorton B, Glushakow-Smith S, Borger D, Wang Y, Miles L, Belizaire R, Kitamura S, Gritsman K, Shechter D. TTL4 glutamyltransferase is a therapeutic target for NPM1-mutated acute myeloid leukemia. bioRxiv. 2025 April 07; :2025.04.07.647605. Available from: <https://www.biorxiv.org/content/10.1101/2025.04.07.647605v1.full> DOI: 10.1101/2025.04.07.647605
- Lorton BM, Warren C, Ilyas H, Nandigrami P, Hegde S, Cahill S, Lehman SM, Shabanowitz J, Hunt DF, Fiser A, Cowburn D, Shechter D. Glutamylation of Npm2 and Nap1 acidic disordered regions increases DNA mimicry and histone chaperone efficiency. iScience. 2024 Apr 19;27(4):109458. PubMed Central PMCID: [PMC10987829](https://pubmed.ncbi.nlm.nih.gov/PMC10987829/).
- Warren C, Matsui T, Karp JM, Onikubo T, Cahill S, Brenowitz M, Cowburn D, Girvin M, Shechter D. Dynamic intramolecular regulation of the histone chaperone nucleoplasmin controls histone binding and release. Nat Commun. 2017 Dec 20;8(1):2215. PubMed Central PMCID: [PMC5738438](https://pubmed.ncbi.nlm.nih.gov/PMC5738438/).
- Warren C, Shechter D. Fly Fishing for Histones: Catch and Release by Histone Chaperone Intrinsically Disordered Regions and Acidic Stretches. J Mol Biol. 2017 Aug 4;429(16):2401-2426. PubMed Central PMCID: [PMC5544577](https://pubmed.ncbi.nlm.nih.gov/PMC5544577/).
- Onikubo T, Nicklay J.J., Xing L., Warren C., Anson B., Wang W.-L., Burgos E.S., Ruff S.E., Shabanowitz J., Cheng R.H., Hunt D.F., Shechter D.. Developmentally regulated post-translational modification of nucleoplasmin controls histone sequestration and deposition. Cell Reports. 2015; 10(10):1735-1748. DOI: 10.1016/j.celrep.2015.02.038

Other Significant Products Highlighting Contributions to Science

- Roth JS, DeAngelo JD, Young DL, Maron MI, Saha A, Pinto H, Gupta V, Jacobs N, Hegde S, Aguilan JT, Basken J, Azofeifa J, Query CC, Sidoli S, Skoultchi AI, Shechter D. PRMT5 activity sustains histone production to maintain genome integrity. bioRxiv. 2025 Jul 7; PubMed Central PMCID: [PMC12265569](https://pubmed.ncbi.nlm.nih.gov/PMC12265569/).
- DeAngelo JD, Maron MI, Roth JS, Hegde S, Silverstein AM, Gupta V, Stransky S, Basken J, Azofeifa J, Query CC, Sidoli S, Gamble MJ, Shechter D. Productive mRNA chromatin escape is promoted by PRMT5 activity. Mol Cell. 2025 Nov 6;85(21):4016-4031.e9. PubMed Central PMCID: [PMC12744693](https://pubmed.ncbi.nlm.nih.gov/PMC12744693/).
- Maron MI, Casill AD, Gupta V, Roth JS, Sidoli S, Query CC, Gamble MJ, Shechter D. Type I and II PRMTs inversely regulate post-transcriptional intron detention through Sm and CHTOP methylation. Elife. 2022 Jan 5;11 PubMed Central PMCID: [PMC8765754](https://pubmed.ncbi.nlm.nih.gov/PMC8765754/).
- Shechter D., Dormann H.L., Allis C.D., Hake S.B.. Extraction, purification and analysis of histones. Nature Protocols. 2007;

2(6):1445-1457. DOI: 10.1038/nprot.2007.202

5. Shechter D, Costanzo V, Gautier J. ATR and ATM regulate the timing of DNA replication origin firing. Nat Cell Biol. 2004 Jul;6(7):648-55. PubMed PMID: [15220931](#).

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

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NIH BIOGRAPHICAL SKETCH SUPPLEMENT

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Personal Statement

My research investigates how post-translational modifications (PTMs) regulate chromatin, transcription, and RNA processing, with a mechanistic focus on how these modifications tune intrinsically disordered regions in nuclear proteins. By integrating biochemical and enzymological studies, structural analysis, and cellular and genome-wide approaches, my lab defines causal relationships between PTMs, protein function, and gene regulation. **A central focus of this work is glutamate-glutamylolation, particularly TTLL4-mediated modification of histone chaperones such as NPM1**, which we have shown alters chromatin association and transcriptional outputs in both normal and disease contexts.

I am committed to rigorous training through structured mentorship, regular lab meetings, and active community engagement, including cofounding the Einstein Chromatin Club and New York City Chromatin Club (NYCCC) and organizing the 2025 in-person NYCCC Symposium. As Chair of the Graduate Division Curriculum Committee and Director of the annual Gene Expression graduate course, I emphasize critical thinking, scientific rigor, and ethical practice. I work closely with each trainee on experimental design, data management, writing, peer review, and career planning, and continue to advance my mentorship through institutional and national programs, including **CIMER (training obtained in 2024)**.

Honors

2022	Julius Marmur Mentorship Award, Einstein's Graduate Programs in the Biomedical Sciences
2018	Julius Marmur Mentorship Award, Einstein's Graduate Programs in the Biomedical Sciences
2018	LaDone H. Schulman Award for Excellence in Teaching, Albert Einstein College of Medicine
2004	PhD with Distinction (top 10%), Columbia University

Contributions to Science

1. Regulation of Histone Chaperones by Intrinsic Disorder and Post-translational Modifications:

My lab has established a mechanistic foundation for understanding how intrinsically disordered, acidic domains in histone chaperones are regulated by post-translational modification, with a focus on how these modifications tune histone binding and chromatin engagement. Using biochemical and biophysical approaches, including NMR spectroscopy, advanced structural modeling, quantitative enzymology, and chromatin assays - we defined how phosphorylation, arginine methylation, and glutamylation modulate the structural ensemble and activity of histone chaperones, initially using embryonic Nucleoplasmin (Npm2) as a tractable model system. This work provided the first direct evidence that PTMs can tune the functional output of acidic disordered regions and established glutamylation as a regulatory modification in histone chaperones (**Onikubo et al., 2015; Warren et al., 2017; Lorton et al., 2024**).

Extending these principles to disease-relevant chromatin biology, we discovered that the glutamyltransferase **TTLL4 modifies NPM1** and other acidic nuclear proteins (Nap1, PTMA), revealing a broader regulatory axis controlling chromatin organization. We showed that TTLL4-dependent glutamylation alters chromatin association, accessibility, and oncogenic transcriptional programs in NPM1-mutant acute myeloid leukemia (AML). Genetic depletion or chemical inhibition of TTLL4 suppresses leukemia cell growth (**Schurer et al., 2025, *bioRxiv* preprint**), identifying TTLL4 as a mechanistically defined and druggable cellular regulator. Together, these findings provide the direct conceptual and experimental foundation for the proposed studies on TTLL4 enzymatic specificity and NPM1 glutamylation-dependent gene regulation.

2. Molecular Mechanisms and Biological Function of Protein Arginine Methylation in Transcription and Splicing:

My research has defined how Protein Arginine Methyltransferases (PRMTs) regulate chromatin dynamics, transcription, and RNA processing, providing a mechanistic foundation for understanding the molecular logic of post-translational modifications on charged, basic IDRs. My group first solved the structure of the conserved eukaryotic PRMT5-MEP50 complex (Ho et al., 2013), establishing a structural paradigm for the field, and developed ultrasensitive enzymological assays that enabled quantitative analysis of PRMT substrate recognition and activity (Burgos et al., 2017). Using these tools, we demonstrated that PRMT5-catalyzed arginine methylation can both activate and repress transcription (Chen et al., 2017; Maron et al., 2021) and regulates post-transcriptional intron detention (Maron et al., 2022). We further identified novel pathways controlling transcript retention on chromatin and defined a new class of RNAs, GRIPPs (Genomically Retained Incompletely Processed Polyadenylated transcripts), linking arginine methylation directly to chromatin organization and gene expression (DeAngelo et al., 2025).

In parallel, I have contributed widely cited conceptual frameworks through reviews on PRMT5 function (Stopa et al., 2015), the cellular consequences of arginine methylation (Lorton and Shechter, 2020), and the roles of WD-repeat proteins in genome regulation (Lorton and Shechter, 2024), helping to shape current understanding of PTM-regulated genome dynamics that underpins this proposal.

3. Studies of Checkpoint Signaling as a Normal Consequence of DNA Replication:

Prior to my independent appointment, my work focused on elucidating how DNA damage checkpoints, primarily mediated by the ATM and ATR kinases, regulate the cell cycle in response to DNA damage. Our early studies were the first to demonstrate that ATM and ATR not only signal in response to exogenous DNA damage but also function during normal cell cycles, providing real-time feedback on replication dynamics and influence the utilization of multiple DNA replication origins. By dissecting how these checkpoints operate on both naked DNA and chromatin templates, we contributed to an improved mechanistic understanding of how replication-associated processes are coordinated with cell cycle progression.

4. Early Embryo Histone Code – Unveiling Chromatin States during Development to Inform PTM-Regulated Genome Function:

Histone post-translational modifications were hypothesized to form a “code” that directs the usage of DNA. I exploited the unique biological context provided by *Xenopus laevis* eggs and embryos - systems naturally primed for rapid cell division and defined by a distinct epigenetic state of maternal control over the zygotic genome - to dissect how chromatin states govern developmental processes and maintain pluripotency prior to differentiation. By characterizing the dynamics of histone variants and modifications during key developmental transitions, my work has substantially advanced our understanding of the histone code in development.

5. Studies of the Replicative Helicase in Archaea and Eukarya:

My early publications focused on identification of the helicase responsible for melting DNA at the DNA replication fork. The nature and function of the putative ATP-dependent enzyme responsible for unwinding the DNA at the replication origin and during ongoing replication had been the subject of some controversy and its identity was unclear. My work was among the very first supporting the function of the MCM (Minichromosome Maintenance) proteins in functioning as an ATP-dependent helicase in Archaea and in frog egg extract. This work is still cited and formed the basis for a substantial subsequent literature on larger assemblies that function to melt the DNA during replication in vivo.

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